

Chapter 1, Problem 8PP

Problem

If an electron beam in a TV picture tube carries 10^{13} electrons/second and is passing through plates maintained at a potential difference of 30 kV, calculate the power in the beam.

Step-by-step solution

Step 1 of 3

Given a TV picture tube carries 10^{13} electrons/second

$$\frac{dn}{dt} = 10^{13} \text{ electrons/second}$$

$$V_o = 30 \text{ kV}$$

The charge on an electron is

$$e = -1.6 \times 10^{-19} \text{ C}$$

Step 2 of 3

If the number of electrons is n , then $q = ne$ and

$$i = \frac{dq}{dt}$$

$$i = e \frac{dn}{dt}$$

$$i = (-1.6 \times 10^{-19})(10^{13})$$

$$i = -1.6 \times 10^{-6} \text{ A}$$

Step 3 of 3

The negative sign indicates that the current flows in a direction opposite to electron flow.

The power in the beam is,

$$p = V_o i$$

$$p = 30 \times 10^3 \times 1.6 \times 10^{-6}$$

$$\therefore \boxed{p = 48 \text{ mW}}$$